

I'm a forward-deployed engineer: I embed with the customer, find the real problem, and ship the AI system to production myself.

\$40B/YR

HP AMPLIFY channel program · \$15B growth target

0 → 1

Products taken from discovery to Tier-1 deployment

\$300M/YR

Nielsen platform shipped; I wrote its revenue case

STATUS KEY · **LIVE** in production · **PILOTS** with customers · **BUILT** running code · **SPECIFIED** written, not built · **IN BUILD** code in progress · **DESIGNED** design complete · **SCOPED** engagement scoped

THE FDE LOOP TWO CASES

EMBED	DISCOVER	ARCHITECT	BUILD	SHIP	GOVERN · MEASURE
CASE 1 · DEFENSE PRIME · SUPPLIER RISK					
Join the account Into the account team for a named account under evaluation. No quota.	The real blocker Not features: the prime won't let an outside party map its product data.	Work inside the limit Prime keeps ownership; the platform holds only the resolution layer (canonical IDs). Query in place.	Demo on their question Configure the shared platform against the buyer's real question. No fork.	Evaluation unblocked Account team owns the close; I carry no quota and commit to no terms.	Post-sale triage Configure → generalize → fork only as last resort, logged with its recurring cost and signed off.
CASE 2 · DESS · A PRIVACY DECISION ON LIVE LP DATA					
Inside the partnership Embedded with the partners who use it daily. [who]	The real blocker LP data under regulator-grade scrutiny. [who needed to see what; who must not]	Configure or generalize? [the request] → [configure / generalize, and why]	Tiers at the query layer Visible / redacted / hidden; an allowlist of revealed fields, never "all except."	Shipped Live on real LP data. [date]	Governed Hashed audit, field-level diffs; mapped to NIST AI RMF · SR 11-7. [metric]

FDE WORK: ONE LIVE, ONE IN PILOTS

BUILT SOLO · DESS IN PRODUCTION · EGON IN PILOTS

DESS

Voice CRM + warm-intro engine

Stealth investor network · live LP data

IN PRODUCTION · LIVE LP DATA

THE CUSTOMER PROBLEM

- Traditional CRMs fail at the moment of capture. Partners don't type notes after a meeting, so the data goes stale and the tool quietly dies.
- "Who do we know who knows this LP?" was tribal knowledge. You couldn't query it.
- Sensitive LP data: regulator-grade scrutiny from day one.

WHAT I SHIPPED

- Voice capture on iPhone (TestFlight): speak it, get a structured, access-controlled record.
- Warm-intro engine: deterministic path scoring, affinity × connection + hops, ≤2 hops, blocklist + consent gates the model cannot override.
- Policy engine on every tool call: deny-by-default, fail-closed; destructive ops need a signed, expiring confirm token.
- PII core: visible / redacted / hidden tiers at the query layer; hashed audit with field-level diffs; mapped to NIST AI RMF · SR 11-7.
- AVA thesis testing (pass · conditional · reject, citation-first) and ARA sealed sharing: answers plus cryptographic receipts, never records.

PARTNER
speaks on iPhone

SPEECH → TEXT
swappable ASR

CLAUDE TOOL-USE
bounded loop · input sandboxed

POLICY GATE
schema · allowlist · confirm

DETERMINISTIC CORE
PostgreSQL · path scoring · privacy tiers

ANSWER
intro path + audit row

Voice is the interface to the data, not a phone line. No LLM in the trust path: the model narrates; code decides who is reachable and who can see what.

EGON

Multi-agent decision support

Extreme-weather risk for the assets defense depends on

PILOTS · 4 COUNTRIES · PRE-PRODUCTION

THE CUSTOMER PROBLEM

- Port authorities, the Army Corps and infrastructure investors need to know what extreme weather will cost a specific asset, what protects it, what that protection is worth, and who else benefits.
- Today's answer: one-off consulting reports with figures no one can trace to a source.

PROVING GROUNDS

- Baltimore Harbor (USACE proof of concept) · Mare Island shipyard, Vallejo

WHAT I BUILT

- 20+ data loaders over federal and science APIs: NOAA CO-OPS · OpenFEMA · USGS · ERDDAP · FRED · Census · NASA CMR.
- Google ADK coordinator fanning out to 10 worker agent classes over A2A: validated envelopes, circuit breaker, retries.
- Provenance by construction: frozen Pydantic blocks. No citation, no figure; computed ⇒ calc trace; projected ⇒ model ref. Deterministic compile → lint → render.
- Financing precedents + who else benefits: each precedent cites the deal it came from (program, cost share, year). Local-economy VIPs are mapped from the largest employers, the investors on their cap tables, and public data. Precedents, not advice.

SITE SPEC
declarative asset + scope

FEDERAL DATA
20+ loaders · registry

AGENT TEAM
ADK coordinator → 10 workers · A2A

PAYLOAD FACTORY
no citation, no figure

COMPILE · LINT · RENDER
deterministic

SCENARIO
cost + protection value · financing precedents · who else benefits

Referee, not advocate. A defensible scenario a human carries into their own decision, never a decision the machine makes.

THE STACK WHAT I BUILD WITH

OWNED, SOLO	Discovery · Architecture · Full stack · ML ops · Multi-tenant cloud · Security · Governance · Rollout
LANGUAGES	Python · TypeScript · Java · SQL
LLMS · AGENTS	Claude · Llama (local) · Claude Code · Google ADK · A2A · MCP · LangGraph
CONTRACTS · TESTS	Pydantic v2 (frozen models) · mpyy --strict · pytest · Playwright · promptfoo
DATA	PostgreSQL/pgvector · SQLAlchemy · BigQuery · rdflib (SKOS/OWL) · Kafka
CLOUD · APP	GCP (Cloud Run · Cloud SQL · Vertex AI) · Confidential Space · AWS · FastAPI · React · Capacitor iOS
VOICE · TRACING	Google Chirp · ElevenLabs · Deepgram · Groq · Arize · OpenTelemetry

HOW I BUILD ENFORCED CONTRACTS · LIGHT DECLARATIVE ORCHESTRATION

REQUEST
thin graph
no LLM
planner

DETERMINISTIC
records prices paths permissions figures ▶ TYPED TOOL ▶ AS-IS ✓

NON-DETERMINISTIC
summaries extraction explanations ▶ LLM STEP ▶ SCHEMA ✓

PROVENANCE BY CONSTRUCTION EGO N
\$2.4M/yr no source ▶ X CAN'T BE BUILT
flood map dead link ▶ I VISIBLE GAP
19,500 sq mi USGS 03086000 ▶ ✓ FIGURE

Governance as code: an architecture manifest generates the diagrams, docs and stack · 39 ADRs with decision + retirement logs.

PRODUCTION PATTERNS AND WHERE EACH ONE RUNS

EVALS · TESTING GRADED BY CODE WHEREVER POSSIBLE · A MODEL ONLY WHERE IT MUST

DESS PROMPTFOO banned-term + injection cases ▶ DETERMINISTIC ASSERTS + LLM rubric ▶ PYTEST mostly negative tests ▶ PLAYWRIGHT end-to-end ▶ MYPY --STRICT types as contracts

EGON BUILT VERBATIM CLAIMS claim text c source text BUILT REFUSAL, BOTH HALVES must refuse ✓ · must not refuse ✓

SPECIFIED ARITHMETIC INVARIANTS combined benefit ≤ do-nothing loss · parts + interaction = whole SPECIFIED ENGINE FAITHFULNESS engine = rendered = narrated, same order

PROMPT-INJECTION DEFENSE DESS

INJECTED INSTRUCTION ▶ PRIVACY FILTER on every read: sees only what you can ▶ FIELD ALLOWLIST reveals are named fields, never "all except" ▶ HMAC TOKEN delete · transfer need your spoken yes ▶ CAPS OFF-SCHEMA 5-step loop; export ceiling server-side ▶ NO SEND TOOL sending takes 2 human clicks

ACCESS · PRIVACY BOTH

DESS OAUTH (team) · EMAIL (others) allowlisted identity ▶ JWT short-lived ▶ visible redacted hidden

EGON X CUSTOMER FINANCIAL RECORDS X PERSONAL IDS IN RESEARCH ESCROW STAYS LOCAL

ONTOLOGY · REUSE FIRST EGO N

BFO ISO/IEC 21838-2 top level ▶ CCO defense + intel mid-level ▶ SKOS · FIBO · DEFENSE · NAICS subjects · capital · programs · industries ▶ CONNECTIVE TISSUE the only layer authored here

ONE REGISTER every borrowed term names INHERIT REFUSE BINDS TO · binds to: none ⇒ a claim, not a fact · test-enforced

GOVERNING CODING AGENTS BESPOKE TOOLING FOR REGULATED, DATA-INTENSIVE BUILDS

VERIFIER BUILT

LOCAL HOOK on commit · advisory CI RUN whole pushed range · authoritative

VERIFIER plan cited ⇌ actual diff ✓ MERGE X BLOCK

WATCHTOWER · OUT OF THE AGENT'S REACH IN BUILD

AGENT CAN WRITE CODING AGENT main

NO CREDENTIAL AGENT CANNOT TOUCH polls main #0412 ✓ #0413 ✓ #0414 X alarm APPEND-ONLY

13 bypass classes in the threat model

FIELD RECORD CUSTOMER-FACING DEPLOYMENTS

2018–21	2021–23	2023–24	2024	2025–26	2026–	2026–
HP	Nielsen	Ripley Decision Advantage	SPINF AI	Google	DESS	EGON
HP API-first channel exchange \$40B/YR	Nielsen / Gracenote Data platform, uninstalled \$300M/YR	Ripley Decision Advantage Compound AI, shipped solo	SpinF AI · BootstrapLabs Voice AI on live sales calls 0 → 1	Alphabet AI engineer	DESS · stealth investor network Voice CRM on live LP data LIVE	Ripley · EGON Decision support · ports PILOTS

Q WHAT'S UNDER THE VOICE INTERFACE?

DESS



Q AN EXTERNAL SERVICE IS SLOW OR FAILING?

Never a 30-second spinner. Never stale data shown as fresh.



No retry on a bad 400 · idempotency keys on writes · long jobs go async: job ID, then poll or push.

Q HOW DO YOU STOP RUNAWAY TOOL LOOPS?

The graph decides what runs next. The model only fills in parameters.



A limit hit stops the run and escalates to a human, with the full trace.

Q THE SYSTEM GAVE A WRONG ANSWER. NOW WHAT?

Classify the failure first. Each cause has a different fix.



Q AUTHENTICATION AND AUTHORIZATION?

Identity is short-lived; access is checked on every request and logged.



Services: workload identity, mTLS, rotated secrets. DESS Google OAuth allowlist for the team, email sign-in for other users -> short-lived JWT; privacy tiers at the query layer.

Q HOW HANDS-ON ARE YOU?

I code daily, Python first. Don't take my word for it:

github.com/leighanne77/DESS

Public tree of the DESS CRM + intro engine. Runs end-to-end on fictional data.

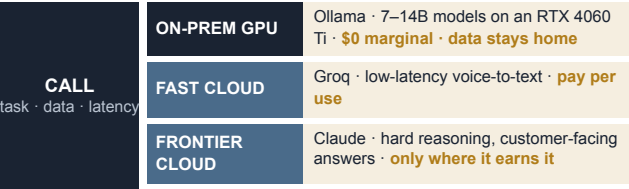
github.com/leighanne77/MAS_Example

Provenance by construction: the EGON trust-core pattern.

```
$ git clone ... && pytest · FastAPI · LangGraph · ADK · Node · Java
```

Q HOW DO YOU MANAGE MODEL COSTS?

Route each call to the cheapest place that meets the bar, and meter everything.



On-prem runs coding assist, the local Llama judge and bulk jobs. DESS per-user token + voice budgets; web-search spend capped and audited. Right-size infra too: PostgreSQL, not Spark.